Section - 8

220 KV, 132 KV SF6 GAS CIRCUIT BREAKERS AND 33 KV VACUUM CIRCUIT BREAKERS

1.0 SCOPE: Refer Clause-1 of Section-I, General

2.0 STANDARDS:

2.1 The circuit breakers shall conform to the latest revisions with amendments available of relevant standards, rules and codes, some of which are listed herein for ready reference.

Sl.No.	Standard	Title
1.	* IEC 62271-100 Ed.1.0 (2001)	Specification for alternating current circuit breakers.
2.	IEC-376	Specification for acceptance of new supply of sulphur hexafloride.
3.	IS-5578 & IS-11353	Making and arrangement for switch gear bus-bar, main connections and auxiliary wiring.
4.	IS-2147	Degree of protection provided for enclosures for low voltage switch gear and control gear.
5.	IEC-71	Electrical Clearances.
6.	IEC-694	Common clauses for high-voltage switch gear and control gear Standards
7.	IS-325	Specification for Induction Motors
8.	IS-2516	Specification for circuit breaker.
9.	IS-2629	Recommended practice for hot dip galvanizing of iron and steel.
10.	IEC-137	Bushings for AC voltages.
11.	IS-2099	High Voltage porcelain bushings.
12.	IS-4379	Identification of the contents of Industrial Gas Cylinders.
13.	IS-7285	Seamless high carbon steel cylinders for permanent and high pressure liquifiable gases.
14.	IS-802	Code of practice for use of structural steel (Part-I) in overhead transmission line towers.

REFERENCE STANDARDS	NAME AND ADDRESS
IEC	International Electro Technical commission Bureau Central de la Commission, Electro Technique International, 1 Rue de verembe, Grneva, Switzerland.
ISO	International Organisations for Standardization Danish Board of Standardization, Danish Standar-dising sraat, Aurehoegvaj - 12 DK - 2900, Heelprup DENMARK
ISS	Indian Standard Bureau of India Standards manak Bhavan,9, Bahadur shah Zafar Marg New Delhi - 110 002 INDIA.

2.2 Equipment meeting with the requirements of any other authoritative standards, which ensure equal or better quality than the standard mentioned above shall also be acceptable, if the equipment offered by the Bidder conforms to any other standards, salient points of difference between the standards adopted and the specific standards shall be clearly brought out in relevant schedule. Four copies of such standards with authentic English translations, shall be furnished along with the offer.

3.0 SERVICE CONDITIONS:

3.1 Climatic Conditions: Refer Clause-2 of Section-I, General

3.2 Auxiliary Power Supply:

Auxiliary electrical equipment shall be suitable for operation on the following supply system.

a. Power Devices (like drive motors)	415 V, 3 phase 4 wire 50 Hz. neutral grounded AC supply.
b. DC supervision control and protective devices	220 V DC, ungrounded 2 wire (Details shall be furnished to the successful bidder).
c. Lighting	240 V, single phase 50 Hz AC supply. Exclusive protection shall provide for space heater, lighting.

Each of the foregoing supplies shall be made available by the Purchaser at the terminal point for each circuit breaker for operation of accessories and auxiliary equipment. Bidder's scope include supply of interconnecting cables, terminal boxes etc., The above supply voltage may vary as indicated below and all devices shall be suitable for continuous operation over the entire range of voltages.

i) AC supply : Voltage + 10% / - 15%;
Frequency +/- 5%

A C Supply shall also have a supervision relay.

ii) DC supply : -15% to + 10%.

4.0 PRINCIPAL PARAMETERS:

The breakers shall conform to the specific technical requirements specified hereunder:

Sl. No.	Item	REQUIREMENTS		
		220kV SF6 CB	132kV SF6 CB	33kV VCB
1.	Rated system voltage (kV rms), Frequency	220 (50 Hz)(+/-5%)	132 (50 Hz)(+/-5%)	33 (50 Hz)(+/-5%)
2.	Maximum System Voltage kV rms	245	145	36
3.	Continuous current rating (A) rms	3150	3150	2500
4.	Type	3 pole outdoor SF6	3 pole outdoor SF6	3 pole outdoor Vacuum
5.	Mounting	Hot dip galvanized lattice steel support structure, bolted type	Hot dip galvanized lattice steel support structure, bolted type	Hot dip galvanized lattice steel support structure, bolted type
6.	Number of Poles	3	3	3
7.	Type of operation	Individually operated single poles	3 pole gang operated	3 pole gang operated
8.	Phase to phase spacing in the switchyard i.e. inter-pole spacing for breaker (mm).	4500	1700	400
9.	Required ground clearance from the lowest live terminal if both the erminals are not in the same horizontal plane. (mm)	5500	4600	3700
10.	Height of concrete plinth	300	300	300
11.	Minimum height of the lowest part of the support insulator from ground level (mm).	2550	2550	2550
12.	Operating mechanism	Spring	Spring	Spring
13.	Auto re-closing duty	Single & Three phase	Three phase	Three phase
14.	Rated operating duty cycle min.	O-0.3 sec.-CO-3min-CO	O-0.3 sec.-CO-3min-CO	O-0.3 sec.-CO-3min-CO
15.	"First pole to clear" factor (type of tripping)	1.3 trip free	1.3	1.5
16.	Max. closing time (ms)	100	100	100

17.	Max. total break time (ms)	Less than 3 cycles or 60 ms	Less than 3 cycles or 60 ms	Less than 3 cycles or 60 ms
18.	1.2/50 microsecond impulse withstand voltage: to earth (kVp)	1050	650	170
19.	1 minute power frequency withstand voltage (kV rms)	460	275	70
20.	Rated breaking current capacity i) Line charging at rated voltage at 90 deg. leading power factor (A) rms. ii) Small inductive current (A) rms without switching o/v exceeding 2.0 p.u. iii) Cable charging current (A) iv) Short circuit current a) AC component (kA rms) b) % DC component c) duration of short circuit in sec.	125 0.5 to 10 250 50 corresponding to minimum opening time as per IEC-56. 3sec	50 0.5 to 10 160 40 corresponding to minimum opening time as per IEC-56. 3sec	---- ----- 31.5 corresponding to minimum opening time as per IEC-56. 3sec
21.	Rated short circuit making current capacity (kA)	125	100	78.75
22.	Permissible limit of temperature Rise	As per clause 5.28 of this Specification	As per clause 5.28 of this Specification	As per clause 5.28 of this Specification
23.	Max. acceptable difference in the instants of closing/opening of contacts i) Within a pole (ms) ii) Between poles (ms)	5 10	5 10	

24.	Min. creepage distance of support insulator (mm)	7595 (31mm)	4495 (31mm)	1116 (31mm)
25.	Short time current carrying capability for one second (kA)	50	40	31.5
26.	i) Rating of auxiliary contacts ii) No. of auxiliary contacts	10 A at 220 V DC 10 NO and 10 NC as spare	10 A at 220 V DC 10 NO and 10 NC as spare	10 A at 220 V DC 10 NO and 10 NC as spare
	iii) Rating of Spring charge limit switch auxiliary contacts	10 A at 220 V DC	10 A at 220 V DC	10 A at 220 V DC
27	Breaking capacity of auxiliary with the contacts circuit time constant not less than 20 ms.	2 A DC	2 A DC	2 A DC
28.	Noise level at base and upto 50 meters	75 db (max.)	75 db (max.)	75 db (max.)
29.	Seismic acceleration (horizontal)	0.3 g	0.3 g	0.3 g
		0.5g horizontal for the site location under the Zone-V as per IS-1893 0.3g horizontal for the site location under other than the Zone-V as per IS-1893		
30.	Min. Corona extinction voltage (kV rms)	156	92	-
31.	No. of closing and tripping coils.	Two trip coils and one close coil with Anti-pump-ing arrangement. (two trip coils along with plungers must be mounted separately)	Two trip coils and one close coil with Anti-pump-ing arrangement. (two trip coils along with plungers must be mounted separately)	Two trip coils and one close coil with Anti-pump-ing arrangement. (two trip coils along with plungers must be mounted separately)
32	Capacitor tripping device	Separate CTD's for DC source-1 & source-2 for 132kV and 220kV only.		
33	SF6 pressure transducers	Pressure transducers to be provided on all 3 phases for monitoring in SAS for 220kV/132KV Circuit Breakers.		
34	SF6 gas transducers	Shall be powered on 220V DC/AC for 132kV, 220kV		

(Bidders offering lesser clearances for spacing must substantiate the same with adequate information on impulse test withstand capability of equipment).

5.0 TECHNICAL REQUIREMENTS:

5.1 220 kV & 132 kV Circuit breakers offered shall be sulphur hexa-flouride (SF6) type only and 33 kV Circuit breakers shall be vacuum type only.

5.2 Any part of the breakers, especially the removable ones, shall be freely interchangeable without the necessity of any modification at site.

5.3.a) 220 kV SF6 Circuit Breaker shall comprise of three identical single pole units and are meant for single pole re-closure. In this case pole discrepancy special features including relays, timers etc., provided to minimize pole discrepancy and to avoid single phasing shall be highlighted in the bid. Complete circuit breaker with all the items necessary for successful operation shall be supplied, including but not limited to the following:

5.3.b) 132 kV SF6 Circuit Breaker and 33 kV vacuum Circuit Breakers shall comprise of three identical single pole units and are not meant for single pole re-closure. The units for these breakers shall be linked together mechanically. Complete circuit breaker with all the items necessary for successful operation shall be supplied, including but not limited to the following:

5.3.1 Breaker assemblies with bases, support structure for circuit breaker as well as for control cabinet, central control cabinet and foundation bolts for main structure (except concrete foundations), terminals and operating mechanisms are included in the scope of supply.

5.3.2 Compressed SF6 gas, pneumatic/hydraulic systems complete including compressors, tanks, piping, fittings, valves and controls and necessary supports for inter-pole piping for pneumatic and hydraulic systems.

5.3.3 In case of 220 kV circuit breakers one central control cabinet for each breaker, and one control box for each pole shall be provided with all the required electrical devices mounted therein and the necessary terminal blocks for termination of inter pole wiring. The necessary inter pole cabling at site shall be done by the Purchaser based on the schematic, wiring diagram and termination schedule to be supplied by the successful Bidder. The central control box cabinet shall be floor mounting and shall not be fixed to any of the poles. The control box of each pole should be fixed to pole and ladder with supports should be arranged for climbing of engineers for easy access.

5.3.4 Instruments, gauges and other devices for SF6 gas pressure, supervision. All pressure gauges should have NRV to enable replacement of the gauges without draining of gas. The pressure gauges should have green zone/red zone indicating safe operating

5.3.5 All necessary parts to provide a complete and operable circuit breaker installation such as main equipment, terminal, control parts, connectors and other devices, whether specifically called for herein or not.

5.4 The 220 kV SF6 circuit breaker shall be designed for high speed single and three pole re-closing and 132 kV SF6 Circuit breakers and 33kV Vacuum Circuit

breakers shall be designed for high speed three pole re-closing with operating sequence and timing as specified in clause 4.0 "Principal Parameters".

5.5 The support structure of circuit breaker as well as that of control cabinet shall be hot dip galvanized. All other parts can be gray color painted as per shade 631 of IS-5.

5.6 Circuit breaker shall be suitable for hot line washing.

5.7 All breakers shall be supplied with terminal connectors as required..

5.8 The current density adopted for the design of the terminal pads shall in no case exceed the following values. For copper pads - 1.6 A/sq. mm. and others - 1.0 A/sq. mm.

5.9 CONTACTS:

5.9.1 All making and breaking contacts shall be sealed free from atmospheric effects. Contacts shall be designed to have adequate thermal and current carrying capacity for the duty specified and to have a life expectancy so that frequent replacements due to excessive burning will not be necessary. Provision shall be made for rapid dissipation of heat generated by the arc on opening.

5.9.2 Main contacts shall be the first to open and the last to close so that there will be little contact burning and wear.

5.9.3 If arcing contacts are used they shall be the first to close and the last to open. These shall be easily accessible for inspection and replacement. If there are no separately mounted arcing contacts the main contacts shall be easily accessible for inspection and replacements. Tips of arcing contacts and main contacts shall be silver plated or have tungsten alloy tipping.

5.9.4 Any device provided for voltage grading to damp oscillations or to prevent restrict prior to the complete interruption of the circuit or to limit over-voltages on closing shall have a life expectancy comparable to that of the breaker as a whole.

5.9.5 Breakers shall be so designed that when operated within their specified rating, the temperature of each part will be limited to values consistent with a long life of the material used. The temperature shall not exceed that indicated in IEC-56 under specified ambient conditions.

5.9.6 Contacts shall be kept permanently under pressure of SF6 gas. The poles shall be partially filled with SF6 gas at the time of dispatch and only topping upto required pressure shall be permitted at site. The gap between the open contacts shall be such that it can withstand atleast the rated phase to ground voltage continuously at zero gauge pressure of SF6 gas due to its leakage.

5.9.7 Specific guarantee that the SF6 gas will not leak more than 1% per year from the breaker for a minimum period of 10 years from supply or commissioning whichever is later shall be given. If leakage of SF6 gas occurs beyond this limit within 10 years the defects in the breaker shall be attended to by the company free of cost and this shall be done irrespective of the number of leakages during

the period of 10 years. The defect shall be rectified within 7 days from the date of intimation (by Fax, Cable, Telephone) including travel time of the service personnel.SF6 gas leak warranty shall mention on name plate.

If the defects due to which the gas leak occurs, are not attended by the bidder within 7 days, the cost towards the quantum of gas to be refilled due to the defective system, shall be recovered from their running bills or shall be made good by the bidder.

5.9.8. Special guarantee for vacuum bottle/interrupter: (for 33kV VCBs)
The bidder should give a special guarantee for vacuum bottle/interrupter for 5 years from the date of supply or commissioning whichever is later . Free replacement, if fails during five years period.

The bidder would not confirm on Clause 5.9.7 and Clause 5.9.8 shall be treated as non-responsive.

5.10 PORCELAIN HOUSING:

5.10.1 The porcelain housing shall be of single piece construction without any joint or coupling. It shall be made of homogeneous, vitreous porcelain of high mechanical and dielectric strength. Glazing of porcelain shall be uniform brown or dark brown colour with a smooth surface arranged to shed away rain water or condensed water particles (fog), free from blisters, burns and similar other defects.

All insulators shall conform to IEC: 61264 for pressurized hollow column insulators IEC: 62155 (hollow pressurized & unpressurized ceramic & glass insulators), IEC: 60815-Selection/dimensioning of high voltage insulators intended for use in polluted conditions, IEC: 60273 characteristics of index/outdoor post insulators and IEC-233 (for other). All routine and sample tests shall be conducted on the hollow column insulators as per these standards with requirements and procedures modified as under and the test reports from the sub-vendor shall be submitted.

5.11 ADDITIONAL REQUIREMENTS:

5.11.1 a) The circuit breakers shall be single pressure type and single break. The design and construction of the circuit breaker shall be such that there is minimum possibility of gas leakage and entry of moisture. There should not be any condensation of SF6 gas on the internal insulating surface of 220 kV and 132 kV SF6 circuit breakers.

b) All gasket surfaces shall be smooth, straight and reinforced, if necessary to minimize distortion and to make a tight seal. Where the operating rod of the operating mechanism connected to the arc chamber (SF6 media) adequate O-ring seals shall be provided.

c) In the interrupter assembly there shall be an absorbing product box to eliminate SF6 decomposition products and moisture. The material used in the construction of the 220 kV and 132 kV SF6 circuit breakers shall be fully compatible with SF6 gas.

d) Each pole of 220 kV and 132 kV SF6 Circuit Breakers shall form an enclosure filled with SF6 gas independent of the other two poles. The SF6 gas density of each pole shall be monitored and regulated by pressure switches .

e) The SF6 gas density monitor of 220 kV and 132 kV SF6 Circuit Breakers shall be adequately temperature compensated. The density monitor shall meet the following requirements:

i) It shall be possible to dismantle the density monitor for checking/replacement without draining the SF6 gas by using suitable interlocked non-return couplings.

ii) It shall damp the pressure pulsation while filling the gas in service so that the flickering of the pressure switch contacts does not take place.

iii) A pressure indicator shall also be supplied.

iv) The alarm and lockout /trip setting shall be such that:

a) Advance warning can be given for the low pressure below an unacceptable level.

b) After alarm, the breaker shall open if the pressure falls below pre-determined value.

v) Pressure Guage readings and Density Monitor shall be clearly visible to the operator standing on the Ground.

vi) Alarm/lockout levels of SF6 gas should be printed on the name plate.

f) Means for pressure relief shall be provided in the gas chamber of circuit breaker to avoid damages or distortion during occurrence of abnormal pressure increase or shock waves generated by internal electrical faults, if required for the Circuit Breakers design. The position of pressure relief devices shall be so arranged as to minimize danger to the operators in the event of gas or vapour escaping under pressure.

g) Sufficient SF6 gas for 220 kV and 132 kV SF6 Circuit Breakers shall be provided to fill all the circuit breakers installed. In addition to this 20% of the total gas requirement shall be supplied in separate cylinders as spare requirement.

h) Provisions shall be made for attaching an operation analyzer after installation at site to record contact travel, speed and making measurement of operation timings and synchronization of contacts in one pole.

5.12 SULPHUR HEXAFLUORIDE GAS (SF6 GAS) : (220 kV and 132 kV SF6 Circuit Breakers)

a) The SF6 gas shall comply with IEC-376, 376A and 376B and be suitable in all respects for use in the switch gear under the worst operating conditions.

b) The high pressure cylinders in which the SF6 gas is shipped and stored at site shall comply with requirements of the following standards and regulations:

IS:4379 Identification of the contents of industrial gas cylinders.

IS:7285 Seamless high carbon steel cylinders for permanent and high pressure liquifiable gases.

The cylinders shall also meet Indian Boiler Regulations.

c) Test: SF6 gas shall be tested for purity, dew point, breakdown voltage, water contents as per IEC-376, 376A and 376B and test certificates shall be furnished to Purchaser indicating all the tests as per IEC376.

5.13 DUTY REQUIREMENTS:

5.13.1 The circuit breakers shall be totally re-strike free under all duty conditions and shall be capable of performing specified duties without opening resistors.

5.13.2 The circuit breakers shall meet the duty requirements for any type of fault or fault location, for line charging and dropping when used on an effectively grounded system and perform make and break operations as per stipulated duty cycles satisfactorily.

5.13.3 The circuit breakers shall be capable of:

i) Interrupting the steady and transient magnetizing current corresponding to 220 kV and 132 kV class transformers of 100/160 MVA and 50 /80 MVA rating respectively.

ii) Interrupting line charging current as given in clause 4.0, "Principal Parameters" of this specification with a temporary over voltage as high as 1.5 p.u. without re-strikes and without use of opening resistors.

iii) Clearing short line faults (Kilometric faults) with source impedance behind the bus, equivalent to symmetrical fault current specified.

iv) Breaking inductive currents of 0.5 to 10 A without switching over voltage exceeding 2.0 p. u.

v) Breaking 25% of the rated fault current at twice rated voltage under phase opposition condition.

5.13.4 The critical current which gives the longest arc duration at lockout pressure of extinguishing medium and the arc duration shall be indicated.

5.13.5 The breakers shall satisfactorily withstand the high stresses imposed on them during fault clearing, load rejection and re-energisation of lines with trapped charges. The breakers shall also withstand the voltage specified in clause 4.0 "Principal Parameters" of the specification.

5.14 TOTAL BREAK TIME:

5.14.1 The "Total Break Time" as specified in clause 4.0, "Principal Parameters" of this section shall not be exceeded under any of the following duties:

i) Test duties 1,2,3,4,5 (with TRV as per IEC-56).
ii) Short line fault L90, L75 (with TRV as per IEC-56).

5.14.2 The bidder may please note that there is only one specified break time of the breaker which shall not be exceeded under any duty conditions specified such as with the combined variation of the trip coil voltage, (70-110%) pneumatic, hydraulic pressure and arc extinguishing medium pressure etc., the proof for the total break time of complete circuit breaker shall be furnished.

5.14.3 The values guaranteed shall be supported with the type test reports.

5.15 OPERATING MECHANISM AND ASSOCIATED EQUIPMENTS:

5.15.1 The circuit breakers shall be designed for electrical local as well as remote control. In addition there shall be provision for local mechanical control (emergency trip).

5.15.2 The operating mechanism for 220 kV SF6 Circuit Breakers shall be pneumatic type or hydraulic type or mechanical spring charging type and where as for 132 kV Sf5 and 33 kV vacuum Circuit Breakers shall be spring charge type operated by electrical control. The mechanism shall be adequately designed for the specified tripping and re-closing duty. The entire operating mechanism control circuitry, individual breaker compressor unit for pneumatically operating mechanism Circuit Breaker, hydraulic pump for hydraulic mechanism Circuit Breaker, spring charge motor for spring mechanism Circuit Breaker etc., as required, shall be housed in an outdoor type, hot dip galvanized or painted steel enclosure.

The enclosure shall be invariably mounted on a separate concrete plinth of 300 height. In case of painted enclosures, the housing shall be painted with shade 631 of IS-5 i.e. epoxy light gray. All interiors and exteriors of the housing shall be thoroughly cleaned to remove all rust, scales, corrosion, grease and other adhering foreign matter and the surface treated by recognized phosphating (i.e. seven tank phosphating sequence). After such preparation of surfaces, two coats of Zinc Oxide primer shall be given by suitable stoving and air drying before final painting. The final painted cubicle shall present aesthetically pleasing appearance free from any dent or uneven surface. Painting inside the metallic housing shall be of anti-condensation type and the paint on outside surface shall be suitable for outdoor installation.

5.15.3 All working parts in the mechanism shall be of corrosion resistant material. All bearings which require greasing, shall be equipped with pressure grease fittings.

5.15.4 The design of the operating mechanism shall be such that it shall be practically maintenance free for a minimum period of ten (10) years from the date of installation under all operating conditions. The guaranteed years of maintenance free operation, the number of full load and full rated short circuit current breaking/operation without requiring any maintenance or overhauling shall be

clearly stated in the bid. As far as possible the need for lubricating the operating mechanism shall be kept to the minimum and eliminating altogether if possible.

5.15.5 The operating mechanism shall be non-pumping (and tripfree) electrically and mechanically/pneumatically under every method of closing. There shall be no rebounds in the mechanism and it shall not require any critical adjustments at site. Operation of the power operated closing device, when the circuit breaker is already closed shall not cause damage to the circuit breaker or endanger the operator.

5.15.6 A mechanical indicator shall be provided to show open and close positions of the breaker. It shall be located in a position where it will be visible to a man standing on the ground level from a distance of 2 meters with the mechanism housing closed and operation counter shall also be provided in the central control cabinet.

5.15.7 The Bidder shall furnish detailed operation and maintenance manual of the mechanism along with operation manual for the circuit breakers.

5.15.8 The cables to the CB mechanism box shall be run through cable trays to be supplied along with the Breaker.

5.16 CONTROL:

5.16.1 The close and trip circuits shall be designed to permit use of momentary contact switches and push buttons.

5.16.2 Each breaker/pole (in case of 220kV/132kV/33kV) shall be provided with two (2) independent tripping circuits and coils each connected to a different set of signals of different DC systems. Provision shall be made for supervision of each trip coil both in open/closed condition of the breaker by wiring necessary NC/NO contacts in series with trip coils. The trip circuit supervision relays provision shall be made in the control/protection panels. The trip coils shall be suitable for trip circuit supervision. The trip coils should have physically separate and independent coils and plungers.

5.16.3 The breaker shall normally be operated by remote electrical control. Electrical tripping shall be performed by shunt trip coils. Each trip coils shall be connected to Separate DC sources (trip coil-I on DC source-I and trip coil-II on DC source-II) and closing coil shall be connected to independent supply with a separate MCB. DC Supply monitoring relays shall be provided for monitoring DC Source-I and DC Source-II. However, provisions shall be made for local electrical control. For this purpose a local/remote selector switch and close and trip push buttons shall be provided in the breaker central control cabinet. Remote located push buttons and indicating lamps shall be provided by the Purchaser.DC supply monitoring relays shall be provided for monitoring DC source-1(TC-1),DC source-2, DC source for close cirvuit and indication circuit.

5.16.4 The trip coils shall be suitable for trip circuit supervision both in open and closed condition. Capacitor tripping device-1&2 in case of DC-1& DC-II supply failure shall be provided. CTD shall operate if both TC1 & TC2 Sources fail. The CTD shall be charged from DC-I. The trip circuit supervision relay would be provided by the Purchaser. Necessary terminals shall be provided in the central control cabinet of the circuit breaker by the Bidder.

5.16.5 Closing coil shall operate correctly at all values of voltage between 85% and 110% of the rated voltage. Shunt trip shall operate correctly under all operating conditions of the circuit breaker upto the rated breaking capacity of the circuit breaker and at all values of supply voltage between 70% and 110% of rated voltage.

5.16.6 Suitable relay for monitoring of DC supply voltage to the control cabinet shall be provided. The pressure switches used for interlock purposes shall have adequate contact ratings to be directly used in the closing and tripping circuits. In case the contacts are not adequately rated and multiplying relays are used then the interlock for closing/opening operation of breaker shall be with "NO" logic of the relay i.e. if the DC supply to the interlock circuit fails then it is essential that breaker trips in order to protect the switch gear and connected equipment like transformers etc., for any possible faults at that time. Provision for automatic tripping of the breaker shall be made along with the flag indication for non-availability of DC supply. The connection must be such that capacitor provided for this purpose gets charged from the DC source provided by the Board only and the capacitor supply is released for the tripping circuit as soon as DC supply fails. It must be possible to close the breaker only after restoration of DC supply. The entire arrangement of CTD (Capacitor Tripping Device) facility shall be mounted on control cubicle for 220 kV and 132 kV SF6 Circuit Breakers and 33kV Vacuum Circuit Breakers.

5.16.7 A local manual closing device which can be easily operated manually by one standing on the ground shall also be provided for maintenance purposes and direction of motion of handle shall be clearly marked for the spring operating mechanism of Circuit Breaker.

5.16.8 The auxiliary switch of the breaker shall be preferably positively driven by the breaker operating rod and where due to construction features, same is not possible a plug-in device shall be provided to simulate the opening and closing operations of circuit breaker for the purpose of testing control circuits.

5.16.9 All Circuit breakers (220KV, 132KV, 33KV) must have dual DC Sources for Trip circuits & monitoring circuits.TC-I & TC-II must be independent of each other with exclusive independent DC sources. Similarly monitoring Circuits of gas, air and Spring Charge should be independent for each trip circuit and closing circuit. There should not be any DC changes over in trip coil & closing Circuits.

5.16.10 The pole discrepancy circuit must be independent for TC-I and TC-II. Trip circuit -1 should have CTD-1 fed from DC-1 of trip circuit-1,Trip circuit-2 should have CTD-2 fed from DC-2 of Trip circuit-2.There should be independent gas monitoring circuits, one for trip circuit-1 and the other for trip circuit-2. AC Circuit for spring may be provided with a supervision relay.

5.16.11 Fail safe logic may be implemented in the entire scheme design.

5.16.12 Two numbers CTDs shall be provided and a fuse Link shall be provided for isolation of Capacitor and for 33kV VCB single CTD is sufficient and provision shall be made to provide 2nd CTD.. Each CTDs shall be a single Unit enclosed in a metal box.

5.16.13 Spring charged indication shall be provided inside control cubicle. Potential free contact of the same shall be provided wired up to TB in cubicle. Separate MCBs shall be provided for each spring charging motor and the rating of MCBs shall be suitably selected to match the starting, running and stalling time. An overload relay shall be provided for protection of the spring charging motor.

5.17 MOTOR COMPRESSED SPRING CHARGING MECHANISM:

Spring operated mechanism shall be complete with motor, opening spring, closing spring and all other necessary accessories to make the mechanism a complete unit. Breaker operation shall be independent of motor which shall be used solely for the purpose of charging the closing spring. Motor rating shall be such that it requires only 15 seconds or less for fully charging the closing spring. Closing operation shall compress the opening spring and keep ready for tripping. The mechanism shall be provided with means for charging the spring by hand. This operation shall be carried out with the doors of the cubicle open. During this process no electrical or mechanical operation of the mechanism shall endanger the operator or damage the equipment. A mechanical indicating device shall be provided to indicate the state of the charge of the spring and shall be visible with the door of the cubicle closed. An alarm shall be provided for indicating spring failing to be charged by a pre-set timer after circuit breaker closing.

The spring mechanism shall be fitted with a local manual release, preferably by a push button to avoid inadvertent operation. Means shall be provided for discharging the spring when the circuit breaker is in the open position without circuit breaker attempting to close.

5.18 PNEUMATICALLY OPERATED MECHANISM:--(Not applicable)

a) Individual compressor unit (ICU) shall be provided for pneumatically operated circuit breakers.

b) The ICU shall be complete with air piping and accessories, all stop valves, tees, pressure reducers etc., required for normal operation of the breaker.

c) The breaker local receivers shall comply with the requirements in clause 5.28.

d) Independently adjustable pressure switches with potential free, ungrounded contacts to actuate a lockout device shall be provided. This lockout device with provision of remote alarm indication shall be incorporated in each circuit breaker to prevent operation whenever the pressure of the operating mechanism is below that of parameter required for satisfactory operation of the circuit breakers. The scheme should permit operation of all blocking and alarm relays as soon as the pressure transient present during the rapid pressure dip has been damped and a reliable pressure measurement can be made. Such facilities shall be provided for following conditions:

i) Trip lockout pressure.

ii) Close lockout pressure.

iii) Auto re-close lockout pressure.

iv) Extreme low pressure.

e) The compressed air mechanism shall be capable of operating the circuit breaker under all duty conditions with the air pressure immediately before operation between 85% and 110% of the rated supply pressure. The make/break time at this supply pressure shall not exceed the specified make/break time within any value of trip coil supply voltage as specified.

f) The compressed air piping shall comply to requirements given in clause 5.28.7.

5.19 HYDRAULICALLY OPERATED MECHANISM: :--(Not applicable)

a) Hydraulically operated mechanism shall comprise of operating unit with power cylinder, control valves, high and low pressure reservoir, motor etc. The electrically driven motor shall comply to the requirements laid down in clause 5.32.

b) The hydraulic oil used shall be fully compatible for the specified temperature range.

c) The oil pressure switch controlling the oil pressure in the high pressure reservoir shall be provided with remote alarm for continuous monitoring of low pressure, high pressure etc., at purchaser's control room. The necessary remote mounting equipment shall be supplied loose.

d) In the event of failure of AC supply to the motor, when the breaker is in closed position, the operating mechanism shall be capable of performing Open - Close - Open operation.

e) The mechanism shall be capable of operating the circuit breaker correctly and performing the duty cycle specified under all conditions with the pressure of hydraulically operated fluid in the operating mechanism at the lowest permissible pressure before make up. The opening time at the lowest pressure for a particular operation shall not exceed the guaranteed operating time within any value of trip coil supply voltage as specified.

f) Trip lockout shall be provided to prevent operation of the circuit breaker below the minimum specified hydraulic pressure. Alarm contacts for loss of Nitrogen shall also be provided.

g) All hydraulic joints shall have no oil leakage under the site conditions and joints shall be tested at factory against oil leakage.

5.20 OPERATING MECHANISM HOUSING:

The operating mechanism housing/control cabinet shall conform to the requirement specified in clause 5.31.

5.21 INTERLOCKS:

It is proposed to electrically interlock the circuit breaker with Purchaser's associated air break isolating switches in accordance with switchyard safety interlocking scheme. All accessories required on breaker side for satisfactory operation of the scheme shall be deemed to be included in the scope of supply of this Specification.

5.22 SUPPORT STRUCTURE:

The Bidder shall include the price of support structure along with the foundation bolts required in the bid proposal sheet (Form–IIIB) and these shall be considered in evaluation. The support structure shall meet the following requirements.

1) The minimum vertical clearance from any energized metal part to the bottom of the circuit breaker (structure) base, where it rests on the foundation pad, shall be as per the principal parameters clause – 4 in this specification.

2) The minimum vertical distance from the bottom of the lowest porcelain part of the bushings, porcelain enclosures or supporting insulators to the bottom of the circuit breaker base, where it rests on the foundation pad shall be 2.55 meters for all voltages.

3) Support structure shall confirm to IS 2062 and IS 806. And the minimum weight of Zinc coating shall be 900 gm/m2 and the minimum coating 63 thickness shall be 125 microns for the breaker to be utilized in the saline area.

4) Foundation bolts inside concrete also shall be supplied by CB manufacturer.

5.23 FITTINGS AND ACCESSORIES:

5.23.1 Following is the major fittings and accessories to be furnished by Bidder in the central control cabinet.

a) Central control cabinet in accordance with clause No.5.31 complete with:

i) Cable glands of double compression type for the following sizes of cables tentatively proposed.

For 33KV/132KV – 25mm – 5 Nos, 22mm – 2 Nos
For 2200 KV – 25mm – 10 Nos. 22mm – 5 Nos

ii) Local/remote changeover switch.
iii) Four digit operation counter shall be provided phase wise for 245kV and for 132/33 KV one counter for CB.
iv) Pneumatic/hydraulic pressure gauges.(Not applicable)
v) Control switches to cut off control power supply.
vi) Fuses and links as required. For DC supply, only Fuses shall be furnished.

vii) The number of terminals provided shall be adequate enough to wire out all contacts and control circuits plus 24 terminals spare for Purchaser's use.
viii) Operation hour meter for motor.
ix) Three pin plug with socket.
x) Illuminating lamp with door switch.
xi) Lock and key for mechanism box.
xii) Capacitor tripping device.
xiii) Thermostatically controlled space heater.
xiv) Antipumping arrangement in the closing circuit.

b) Anti-pumping relay.

c) Rating and diagram plate in accordance with IEC incorporating year of manufacture.

5.23.2 Additional fittings for pneumatically operated circuit breaker.

a) Unit compressed air system in accordance with clause No.5.29.

b) Breaker local air receivers.

c) Pressure gauge spring loaded safety valve, pressure switch with adjustable contacts and explosion vents**.**

d) Pressure switch to initiate an alarm if the pressure in the aux. reservoir remains below a preset level for longer than it is normally necessary to refill the reservoir.

e) Stop, non-return and other control valves, pipings and all accessories upto breaker mechanism housing.

5.24 SURFACE FINISH:

5.24.1 All interiors and exteriors of tanks, control cubicles and other metal parts shall be thoroughly cleaned to remove all rust, scales, corrosion, greases or other adhering foreign matter. All steel surfaces in contact with insulating oil, as far as accessible, shall be painted with not less than two coats of heat resistant, oil insoluble, insulating paint.

5.24.2 All metal surfaces exposed to atmosphere shall be given two primer coats of zinc chromate and two coats of epoxy paint with epoxy base thinner. All metal parts not accessible for painting shall be made of corrosion resisting material. All machine finished or bright surfaces shall be coated with a suitable preventive compound and suitably wrapped or otherwise protected. All paints shall be carefully selected to withstand tropical heat and extremes of weather within the limits specified. The paint shall not scale off or wrinkle or be removed by abrasion due to normal handling. All external paintings shall be as per shade No.631 of IS:5.

5.24.3 All ferrous hardware, exposed to atmosphere, shall be hot dip galvanised.

5.25 GALVANISING:

All ferrous parts including all sizes of nuts, bolts, plain and spring washers, support channels, structures, shall be hot dip galvanised conforming to latest version of IS:2629 or any other equivalent authoritative standard.

5.26 EARTHING:

The operating mechanism housing, control cabinets, dead tanks, support structure etc., shall be provided with two separate earthing terminals suitable for bolted connection to 75x8mm galvanaised MS flat to be provided by the Purchaser for connection to station earth mat.

5.27 NAME AND RATING PLATES:

Circuit breaker and its operating device shall be provided with a rating plate or plates marked with but not limited to following data:

a) Manufacturer's or trade mark.
b) Serial number or type designation making it possible to get all the relevant information from the manufacturer.
c) Year of manufacture.
d) Rated voltage.
e) Rated insulation level.
f) Rated frequency.
g) Rated normal current.
h) Rated short circuit breaking current.
i) First pole to clear factor.
j) Rated duration of short circuit.
k) Rated auxiliary DC supply voltage of closing and opening devices.
l) Rated pressure of compressed air gas for operation and interruption.
m) Rated out of phase breaking current.
n) Rated supply voltage of auxiliary circuits.
o) Rated supply frequency of auxiliary circuits.
p) SF6 pressure setting for alarm
q) SF6 pressure setting for Lockout
r) SF6 pressure setting for trip
s) APTRANSCO
t) PO/LOI No
u) Creepage distance

The coils of operating devices shall have a reference mark permitting the data to be obtained from the manufacturer.

The rating plate shall be visible in position of normal service and installation. The rating plate shall be weather proof and corrosion proof.

5.28 LIMITS OF TEMPERATURE RISE:

The temperature rise on any part of equipment shall not exceed the maximum temperature rise specified below under the conditions specified in test clauses.

Sl. No.	Nature of the part or of the liquid	Maximum value of	
		Temperature	Temperature rise at a max. ambient air temp. not exceeding 50 deg. C.
1.	Contacts in air, silver faced	105	55

	copper, copper alloy or aluminum alloy(see notes (i)&(ii) Bare copper or tinned Al. alloy	75	25
2.	Contacts in oil: Silver faced copper, copper alloy or aluminum alloy (see note-ii) Bare copper or tinned Al. alloy	90 80	40 30
3.	Terminals to be connected to external conductors by screws or bolts silver faced (see note-iii)	105	55
4.	Metal parts acting as springs		(See note iv)
5.	Metal parts in contact with insulation of the following classes: Class Y : (for non-impregnated materials) Class A : (for materials immersed in oil or impregnated) Class E : in air in oil Class B: in air in oil Class F : in air in oil enamel: Oil base Synthetic in air Synthetic in oil	 90 100 120 100 130 100 155 100 100 120 100	 40 90 70 50 80 50 105 50 50 70 50

5.	Metal parts in contact with insulation of the following classes: Class Y : (for non-impregnated materials) Class A : (for materials immersed in oil or impregnated) Class E : in air in oil Class B: in air in oil Class F : in air in oil	 90 100 120 100 130 100 155 100	 40 90 70 50 80 50 105 50

	enamel: Oil base Synthetic in air Synthetic in oil	100 120 100	50 70 50
6.	Any part of metal or of insulating material in contact with oil, except contacts.	100	50
7.	Oil	100	50

Notes: i) When applying the temperature rise of 55 deg.C. care should be taken to ensure that no damage is caused to the surrounding insulating materials**.**

ii) The quality of the silver facing shall be such that a layer of silver remains at the points of contact after the mechanical endurance test. Otherwise, the contacts shall be regarded as "bare".

iii) The values of temperature and temperature rise are valid whether or not the conductor connected to the terminals is silver faced.

iv) The temperature shall not reach a value where the elasticity of the material is impaired. For pure copper, this implies a temperature limit of 75 deg.C.

5.29 COMPRESSED AIR SYSTEM(Not applicable)

5.29.1 Each circuit breaker shall be supplied with individual compressor unit. Each ICU shall be provided with compressor, piping, piping accessories, local air receivers, control valves, filters, coolers of adequate capacity, reducing valves, isolating valves, drain ports etc.,

5.29.2 The capacity of local air receivers shall be sufficient for two CO operations of the circuit breaker, at the lowest pressure for auto re-closing duty without refilling.

5.29.3 AIR COMPRESSOR: (Not applicable)

a) The air compressor shall be of air cooled type complete with cylinder, lubrication, drive motor and slide rails. Each compressor shall be rated for the following duty:

i)	Total running time of compressor per day	Not exceeding 240 minutes.
ii)	Normal running air charging	Not exceeding 15 minutes.
iii)	Air charging time after one close-open operation	Not exceeding 15 minutes.

b) Compressor shall be driven by automatically controlled motors conforming to this specification.

c) The compressor shall be provided with automatic adjustable unloading device during starting.

d) Each compressor shall be equipped with a starting operation counter and a running barometer.

5.29.4 AIR RECEIVERS: (Not applicable)

a) Air receiver shall be designed in accordance with the section VIII of BS : 5197 for pressure vessel .A corrosion allowance of 3.0 mm. shall be provided for shell and dished ends. Receivers shall be coated on the inside face with anti-rust medium.

b) Air receivers shall have a manhole with cover for inspection and cleaning. Connections for air inlet and outlet, drain and relief valves shall be flanged type. Pressure gauge and pressure switch connections shall be of screwed type.

c) Accessories such as suitable sized safety valve to relieve full compressor discharge at a set pressure equal to 1.1 times the maximum operating pressure, blow off valve, auto drain tap with isolating and bypass valve, dial type pressure gauge with isolating and drain valve and test connection shall be provided.

d) The receiver shall be rated such that the compressor will not be required to start within less than four hours for supplying the leakage of air from the compressed air system.

5.29.5 QUALITY OF AIR: (Not applicable)

Compressed air used shall be free of dust particles. Arrangement for conditioning the compressed air shall be provided as an integral part of air compressor system.

5.29.6 CONTROLS AND CONTROL EQUIPMENT: (Not applicable)

a) The compressor control shall be of automatic start/stop type initiated by pressure switches on the main receiver. Supplementary manual control shall also be provided.

b) Duplicate incoming supply of 415 V, AC shall be provided by the Purchaser at switchyard motor control centre.
The compressor MCC which is in Bidder's scope shall be as per para (c) of this section. Manual changeover scheme from the source to other shall be arranged. Facility to annunciate failure of power supply to the compressor motor control centre shall be provided.

c) All control equipment shall be housed in a totally enclosed sheet steel cabinet conforming to relevant standards. Pressure gauges, other indicating devices and control switches shall be mounted on the control cabinet.

5.29.7 COMPRESSED AIR PIPING, VALVES AND FITTINGS: (Not

applicable)

a) The flow capacity of all valves shall be atleast 20% greater than the total compressor capacity.

b) The high pressure pipe and air system shall be such that after one 0-0.3 sec. CO operation the breaker shall be capable of performing one CO-operation within 3 minutes.

c) All compressed air piping shall be bright annealed, seamless phosphorous deoxidized non-arsencial copper alloy or stainless steel pipe.

d) Compressor air piping in the switchyard trenches will be laid with a slope of 1 in 1000. The piping will be laid horizontally on steel supports in RCC trenches having a slope of 1 in 500. All necessary fittings required for this purpose including the drain valves at all the low points shall be included in the proposal.

e) All joints and connections in the piping system shall be brazed or flared as necessary.

f) All compressed air piping shall be carried out in accordance with latest Indian Standard.

g) Compressed air piping system shall be complete with saddle clamps to support the piping system at every one metre interval and expansion joints at every 40 metres interval.

5.29.8 MOTOR CONTROL CENTRE: (Not applicable)

The 415 Volt motor control centres conforming to requirements for control cabinets stipulated in clause 5.31 of this specification shall be fixed type and fully sectionalized and shall be equipped with load break switches. Motor feeders shall be provided with switch fuse unit and contactor with thermal overload relay and open phase protection. The motor contactor shall have one normally open auxiliary contact for alarm purposes.

An ammeter shall be provided on the motor feeders. The motor control circuit shall be independent from all other control circuits.

5.29.9 ISOLATING SWITCHES:

The incoming power supply isolating switch operating handle shall be interlocked with the control cabinet door as to prevent opening of door when main switch is closed. Device for bypassing the door interlock shall also be provided. Switch handle shall have provision for locking in both fully open and fully closed positions.

5.29.10 FUSES:

All fuses shall be of MCB type, conforming to relevant IS. Fuses shall be provided with visible operation indicators to show that they have operated. All accessible live connections shall be adequately shrouded, and it shall be possible to change fuses with the circuit live, without damage of contact with live conductor. Insulated fuses pulling handle shall be supplied with each control cabinet.

5.30 TERMINAL CONNECTORS:

5.30.1 The terminal connectors suitable for Moose(31.77 mm) ACSR conductor 6 Nos. per circuit breaker shall be supplied and shall meet the following requirements.

a) Terminal connectors shall be manufactured and tested as per IS:5561.

b) All castings shall be free from blow holes, surface blisters, cracks and cavities. All sharp edges and corners shall be blurred and rounded off.

c) No part of a clamp shall be less than 10 mm thick.

d) All ferrous parts shall be hot dip galvanised conforming to IS:2633.

e) For bimetallic connectors, copper alloy liner of minimum thickness of 2 mm shall be cast integral with aluminum body.

f) Flexible connectors shall be made from tinned copper/aluminum sheets.

g) All current carrying parts shall be designed and manufactured to have minimum contact resistance.

h) Connectors shall be designed to be corona free in accordance with the requirements stipulated in IS:5561.

i) For connecting ACSR conductors Aluminum alloy casting, conforming to designation A6 of IS: 617 and all test shall conform to IS: 617.

5.31 SPECIFICATION FOR CONTROL CABINETS:

1. Control cabinets shall be of the free standing floor mounting out door type.

2. Control cabinets shall be sheet steel enclosed and shall be dust, weather and vermin proof. Sheet steel shall be atleast 3.0 mm. thick . 15 mm. thick neoprene or better type of gaskets shall be provided to ensure degree of protection of atleast IP 55 as per IS:2147. Control cabinets shall be provided with hinged door and padlocking arrangement. The door hinges shall be of union joint type to facilitate easy removal. Door shall be properly braced to prevent wobbling. There shall be provision in the bottom sheet for entrance of cables from below.

3. Equipment and devices shall be suitable for operation on a 415 V, 3 phase 3/4 wire 50 Hz. AC system. The wiring shall be of stranded copper with minimum cross section of 2.5 sq.mm. with 1100 V PVC insulation.

4. Bus bars shall be of tinned copper of adequate cross-section to carry the normal current, without exceeding the permissible temperature rise over an ambient temperature of 50 deg.C. outside the cubicle. The buses shall be braced to withstand forces corresponding to short circuit current of 10 kA.

5. Motors rated 1 kW and above being controlled from the control cabinet shall be suitable for operation on a 415 V, 3 phase 50 Hz. system. Fractional kW motors shall be suitable for operation on a 240 V, 1-phase, 50 Hz. supply system.

6. Isolating switches shall be group operated units (3-pole for use on 3-phase supply systems and 2 pole for single phase supply systems). Quick make quick break type, capable of breaking safely and without deterioration, the rated current of the associated circuit. Control cabinet door shall be interlocked with the operating handle of the switch so as to prevent opening of the door when the switch is closed. A device for bypassing the door interlock shall also be provided. Switch handle shall have provision for locking in both fully open and fully closed positions.

7. Fuses shall be HRC cartridge link type having prospective current rating of not less than 46 kA (rms). They shall be provided with visible operation indicators to show when they have operated. One fuse pulling handle shall be supplied for every ten fuses or a part thereof. Pluggable type relays ex: PLA are not allowed.

8. Push button shall be rated for not less than 6 Amps. 240 V AC or 2 Amps, 220 V DC and shall be flush mounted on the cabinet door and provided with appropriate name plates. Red, Green and Amber indicating lamps shall be flush mounted and provided with series resistors to eliminate the possibility of short circuiting of control supply in the event of fusing of lamps.

9. For motors upto 5 kW, contactor shall be direct-on-line, air break, single throw type and shall be suitable for making and breaking the starting current of the associated motor which shall be assumed equal to 6.5 times the full load current of the motor at 0.2 p.f. For motors above 5 kW, automatic star delta type starters shall be provided. 3 pole contactor shall be furnished for 3 phase motors and 2 pole contactor for single phase motors. Reversing contactor shall be provided with electrical interlocks between forward and reverse contactor. If possible, mechanical interlocks shall also be provided. contactor shall be suitable for uninterrupted duty and shall be of duty category class AC4 as defined in IS:2959. The main contacts of the contactor shall be silver plated and the insulation class for the coils shall be class E or better. The dropout voltage of the contactor shall not exceed 70% of the rated voltage.

10. Contactor shall be provided with a three element, positive acting, ambient temperature compensated, time lagged, hand reset type thermal overload relay with adjustable setting. Hand reset button shall be flush with the front door of the cabinet and suitable for resetting with starter compartment door closed. Relays shall be either direct connected or CT operated depending on the rated motor current.

11. Single phasing preventor relay shall be provided for 3 phase motors to provide positive protection against single phasing.

12. Mini starters shall be provided with no volt coils wherever required.

13. All necessary cable terminating accessories such as glands, crimp type tinned copper lugs etc., for control cables shall be included in Bidder's scope of supply. Suitable brass cable glands shall be provided for cable entry.

14. Wiring for all control circuits shall be carried out with 1100/650 volts grade PVC insulated tinned copper stranded conductors of sizes not smaller than 2.5 sq.mm. Atleast 10% spare terminal blocks for control wire terminations shall be provided on each panel.

    The terminal blocks shall be ELMEK type or equivalent. All terminals shall be provided with ferrules indelibly marked or numbered and these identifications shall correspond to the designations on the relevant wiring diagrams. The terminals shall be rated for adequate capacity which shall not be less than 10 Amps.

15. Control cabinet shall be provided with 240 V, 1-phase 50 Hz, 20 W fluorescent light fixture and 5 Amps. 3 pin socket for hand lamp.

16. Space heaters shall be provided inside each cabinet complete with thermostat (preferably differential type) to prevent moisture condensation. Heaters shall be controlled by suitably rated double pole miniature circuit breakers.

17. Signal lamps provided shall be of neon screw type with series resistors, enclosed in bake lite body. Each signal lamp shall be provided with a fuse integrally mounted in the lamp body.

18. All AC control equipment shall be suitable for operation on 240 V, 1 phase 50 Hz. system.

19. Items inside the cabinet made of organic material shall be coated with a fungus resistant varnish.

20. All doors, panels, removable covers and breaker openings shall be gasketed all around. All louvers shall have screens and filters. Cabinets shall be dust, moisture and vermin proof.
    The inter-pole cabling must be completed at factory with necessary ferruling, terminal lugs and cable glands.

21. General Comments

1. DC changeover is not allowed.
2. Independent DC with Separate MCBs & DC supervision required for Close, TC-1, TC-2 & Indication Circuits.
3. Incorporate DC supply supervision relays at the end of the TC-1, TC-2, closing & Indication circuits.

4. Independent CTDs required for both TC-1 & TC-2 Circuits.
5. Both CTD out puts should go through Link. This is useful at the time of trouble shooting
6. Provide Each CTD in Separate Closed Box
7. Independent SF6 lock out contacts required for both TC-1 & TC-2 Circuits.
8. In corporate both SF6 lock out contacts in Close circuit.
9. A single phase trip command should cause a single pole trip, irrespective of the position of the Local/Remote switch.
10. Provide AC supply Supervision and AC Fail contact.
11. Provide AR I/L-1 as CB Unhealthy (Parallel all NC Contacts of SF6 Lockout-1, SF6 Lockout-2 & Spring Status of each limb)
12. Provide AR I/L-2 as CB Healthy (Series of all NO Contacts of SF6 Lockout-1, SF6 Lockout-2 & Spring Status of each limb).
13. Provide CB Local/Remote Selection for SCADA.
14. TC-1 & TC-2 Should be Physically and electrically separate. Both Trip coils must have independent plungers.
15. Pole discrepancy Timer Range should be 0.3 - 10 sec
16. R, Y & B Phase mechanisms should be identical and Separate Marshalling box required
17. Breaker Marshalling box receive 3-phase 4 wire AC Supply. A separate incoming ac supply with 4-pole MCB and single phase preventer must be made available. All internal wiring distribution for AC supply must done at the factory with 3 outgoing MCBs for motor supply (for R, Y & B Limbs), A separate single out going MCB for Heating circuit and Illumination circuit for 3 Mechanism boxes and Marshalling box.
18. Use power contactors, Contact Multiplication Relays are not allowed.
19. Inter Pole wiring should be provided by the manufacturer along with ferrules/lugs terminated at both ends and glands.
20. Minimum Spare Auxiliary Direct Switch contacts should be 10NO+10NC
21. All TB's should have separate Group Marker for each group.
22. All equipment should be mounted on "C and/or J" channel only.
23. Only stranded conductor shall be used. Minimum size of the conductor used for control wiring shall be 1.5 sq.mm. (Multi Strands Copper).

24. SF6 pressure gauge should have green and red zones to differentiate safe level of pressure and abnormal level of pressure respectively. The pressure gauge should be located so that person standing on the ground, can clearly see the reading of the pressure gauge.
25. Closing Circuit should involve the following
    a) Closing Circuit
    b) Anti-Pumping Circuit
    c) Spring Charge Circuit
26. Trip Circuit-1 should involve the following
    a) Trip Circuit-1
    b) Pole Discrepancy Circuit-1
    c) SF6 Lockout Circuit-1
    d) CTD Circuit-1
27. Trip Circuit-2 should involve the following
    a. Trip Circuit-2
    b. Pole Discrepancy Circuit-2
    c. SF6 Lockout Circuit-2 d. CTD Circuit-2
28. Indication Circuit should involve the following
    a. Electrical CB Open
    b. Electrical CB Close
    c. Electrical Spring Charge
    d. Mechanical Operation Counter
29. Spare/Auxiliary Circuit should involve the following
    a. All Status's (SF6 Low, SF6 Lockout, Dc Fail Supervision , Ac Fail Supervision, CB Local/ Remote, Spring Charge etc.)
    b. Spare 52 Auxiliary Switch Contacts
    c. Spare Spring Charge Switch Contacts
    d. All Other Spare Contacts
30. AC Circuit should involve the following
    a. Spring Charge Motor Circuit
    b. Illumination and Heating Circuit.
31. SF6 pressure gauge transducer should be provided with output of 4-20mA for monitoring of pressure in SCADA.
32. SF6 gas density monitor switches should have Non Return Valves for easy replacement of switches without draining the SF6 gas.

33. For 33kV VCBs 1No. CTD shall be provided and provision shall be made to provide 2nd CTD.
34. For reference, please verify approved M/s. ABB GTP, GA & Drawing is uploaded in the website.

5.32 MOTORS:

5.32.1 Motors shall be Universal type of sufficient size capable of satisfactory operation for the application and duty as required for the driven equipment.

6.0 **TESTS:**

6.1 Type Tests:

6.1.1 All the equipment offered, shall be fully type tested by the bidder as per the relevant standard. The bidder shall furnish one set of the following type test reports for each equipment offered along with bid.

The type tests must have been conducted not earlier than 10 (ten) years for 220kV, 132kV & 33kV from the date of bid opening as per latest CEA guidelines.

i) Mechanical Endurance Test.
ii) Power frequency voltage withstand tests (wet & dry)
iii) Lightning Impulse withstand test.
iv) Basic short circuit tests, short line fault test duties and out – of phase switching tests.
v) Short time withstand current and peak withstand current tests.
vi) Line and cable charging current tests.
vii) Temperature rise test.
viii) Seismic withstand test.
ix) Capacitive current switching tests.
x) RIV (Radio Interference) Voltage Test. (for 220 kV & 132 kV SF6 Circuit Breakers only)

6.2 ACCEPTANCE AND ROUTINE TESTS:

6.2.1 All acceptance and routine tests as stipulated in the relevant standards shall be carried out by the Supplier in the presence of Purchaser's representative.

6.2.2 In addition to the mechanical and electrical tests specified by IEC, the following shall also be performed.

Speed curves for each breaker shall be obtained with the help of a suitable operation analyzer to determine the breaker contact movement during opening, closing, auto-re-closing and trip free operation under normal as well as limiting operating conditions (control voltage, pneumatic pressure etc.). The tests shall show the speed of contacts directly at various stages of operation, travel of contacts, opening time, closing time, shortest time between separation and meeting of contacts at break/make operation etc. This test shall also be performed at site for which the necessary operation analyzer along with necessary

transducers, cables, console, etc., shall be furnished as mandatory maintenance equipment.

6.2.3 Immediately after finalisation of the programme of type/acceptance/routine testing, the supplier shall give sufficient advance intimation to the Purchaser to enable him to depute his representative for witnessing the tests.

6.3 ADDITIONAL TESTS:

The Purchaser reserves the right for carrying out any other tests of a reasonable nature at the works of the Supplier/laboratory or at any other recognised laboratory/research institute in addition to the above mentioned type, acceptance and routine tests at the cost of the Purchaser to satisfy that the material complies with the intent of this specification.

7.0 INSPECTION:

7.1 The inspection may be carried out by the Purchaser at any stage of manufacture. The Supplier shall grant free access to the Purchaser's representative at a reasonable time when the work is in progress. Inspection and acceptance of any equipments under this specification by the Purchaser shall not relieve the Supplier of his obligation of furnishing equipment in accordance with the specification and shall not prevent subsequent rejection if the equipment is found to be defective.

7.2 The supplier shall keep the Purchaser informed in advance, about the manufacturing programme so that arrangement can be made for inspection.

7.3 The Purchaser reserves the right to insist on witnessing the acceptance/routine testing of the bought out items.

7.4 No material shall be dispatched from its point of manufacture unless the material has been satisfactorily inspected and tested.

7.5 The supplier shall give 15 days (for local supplies)/ 30 days. (In case of foreign supplies) advance, intimation to enable the purchaser to depute his representative for witnessing acceptance and routine tests.

All charges in connection with inspection by purchaser's personal such as travel, accommodation and incidentals shall be borne by the Purchaser.

8.0 QUALITY ASSURANCE PLAN:

8.1 The bidder shall invariably furnish the following information along with his offer failing which the offer shall be liable for rejection. Information shall be separately given for individual type of equipment offered.

i) The Structure of organization
ii) The duties and responsibilities assigned to staff ensuring quality of work

iii) The system of purchasing, taking delivery and verification of materials
iv) The system for ensuring quality of workmanship
v) The quality assurance arrangements shall confirm to the relevant requirement of ISO-9001 on ISO 9002 as applicable.
vi) Statement giving list of important raw materials, names of sub-supplies for the raw materials, list of standards according to which the raw material are tested, list of tests normally carried out on raw material in the presence of suppliers representative, copies of test certificates.
vii) Information and copies of test certificates as on (i) above in respect of bought out items.

viii) List of manufacturing facilities available

ix) Level of automation achieved and list of areas where manual processing exists.

x) List of areas in manufacturing process, where stage inspections are normally carried out for quality control and details of such tests and inspections.

xi) List of testing equipment available with the bidder for final testing of equipment specified and test plant limitation, if any vis-a-vis the type. Special acceptance and routine tests specified in the relevant standards. These limitations shall be very clearly brought out in "Schedule of Deviations" from the specified test requirements.

8.2 The Contractor shall within 30 days of placement of order, submit the following information to the purchaser.

i) List of the raw material as well as bought out accessories and the names of sub-suppliers selected from those furnished along with the offer.

ii) Type test certificates of the raw material and bought out accessories if required by the purchaser.

iii) Quality Assurance Plan (QAP) with hold points for purchasers inspection. QAP and purchasers hold points shall be discussed between the purchaser and contractor before the QAP is finalised.

The contractor shall submit the routine test certificates of bought out accessories and central excise asses for raw material at the time of routine testing if required by the purchaser and ensure that the quality assurance requirements of specification are followed by the sub-contractor.

8.3 The Quality Assurance Programme shall give a description of the Quality System and Quality Plans with the following details. Quality Assurance Plan shall be submitted along with sub-vendor list to Quality wing/ APTRANSCO for prior approval

i) Quality System:

- The Structure of the Organisation
- The duties and responsibilities assigned to staff ensuring quality of work.
- The system of purchasing, taking delivery of verification of materials.

- The system of ensuring of quality workmanship.
- The system of control of documentation.
- The system of retention of records.
- The arrangement of contractor internal auditing.
- A list of Administrator and work procedures required to achieve contractors quality requirements. These procedures shall be made readily available to the purchaser for inspection on request.

ii) Quality Plans:

- An outline of the proposed work and program sequence.
- The structure of contractors organizations for the contract.
- The duties and responsibilities ensuring quality of work
- Hold and notification points.
- Submission of engineering documents required by this specification.
- The Inspection of the materials and components on request.
- Reference to contractors work procedures appropriate to each activity.
- Inspection during fabrication/construction.
- Final inspection and test.

8.4 Training of Personnel:

The contractor shall provide necessary facilities for training personnel at their works/principals works relating to design, manufacture, assembly and testing and operation maintenance for six personnel free of cost. However, travel and incidental charges of the personnel will be borne by the purchaser.

9.0 DOCUMENTATION:

9.1 All drawings shall conform to International Standards Organisation (ISO) 'A' series of drawing sheet/Indian Standards Specification. All drawings shall be in ink and suitable for micro filming. All dimensions and data shall be in S.I. Units.

9.2 List of Drawings and Documents:

The Bidder shall furnish four sets of relevant descriptive and illustrative published literature pamphlets and the following drawings for preliminary study along with the offer.

a) General outline drawings showing dimensions and shipping weights, quantity of insulating media, air receiver capacity etc.,

b) Sectional views showing the general constructional features of the circuit breakers including operating mechanism, arcing chambers, contacts etc., with lifting dimensions for maintenance.

c) Schematic diagrams of breakers offered for control supervision and re-closing.

d) Structural drawing, design calculations and loading data for support structures.

e) Foundation drilling plan and loading data for foundation design.

f) Type test reports.

g) Single line Diagram of SF6 gas circuit shall be incorporated into the drawing.

9.3 The Supplier shall within 2 weeks of placement of order submit four sets of final version of all the above drawings for Purchaser's approval. The Purchaser shall communicate his comments/approval on the drawings to the Supplier within two weeks of receipt of drawings. The supplier shall, if necessary, modify the drawings and resubmit four copies of the modified drawings for Purchaser's approval .The purchaser shall communicate approval with in two weeks of receipt of modified drawings. After receipt of Purchaser's approval, the Supplier shall, within three weeks, submit 20 prints and two good quality re-producibles of the approved drawings for Purchaser's use. After approval of the drawings by APTRANSCO, the supplier can proceed for manufacturing process.

9.4 The Supplier shall also furnish twenty copies of bound manuals covering erection, commissioning, operation and maintenance instructions in English Language and all relevant information and drawings pertaining to the main equipment as well as auxiliary devices. Marked erection drawings shall identify the component parts of the equipment as shipped to enable Purchaser to carryout erection with his own personnel. Each manual shall also contain one set of all the approved drawings, type test reports as well as acceptance reports of the corresponding consignment dispatched.

9.5 The manufacturing of the equipment shall be strictly in accordance with the approved drawings and no deviation shall be permitted without the written approval of the Purchaser. All manufacturing and fabrication work in connection with the equipment prior to the approval of the drawing shall be at the supplier's risk.

9.6 Approval of drawings by the Purchaser shall not relieve the supplier of any of his responsibility and liability for ensuring correctness and correct interpretation of the drawings for meeting the requirements of the latest revision of the applicable standards, rules and codes of practices. The equipment shall conform in all respects to high standards of engineering, design, workmanship and latest revisions of relevant standards at the time of supply and Purchaser shall have the power to reject any material which, in his judgment, is not in full accordance therewith.

9.7 Additional data to be furnished along with the offer:

a) Drawing showing contacts in close, arc initiation, full arcing, arc extinction and open position.

b) The temperature v/s pressure curves for each setting of density monitor, along with details of density monitor.

c) Method of checking the healthiness of voltage distribution devices (condensers) provided across the breaks at site.

d) Data on capabilities of circuit breakers in terms of time and number of operations at duties ranging from 100% fault currents to load currents of the lowest possible value without requiring any maintenance or checks.

e) Sectional view of non return couplings if used for SF6 pipes (for SF6 Circuit Breakers).

f) Details and type of filters used in interrupter assembly and also the operating experience with such filters (for SF6 C.B.)

g) Details on SF6 Gas :

i) Proposed tests to assess the condition of SF6 gas within a circuit breaker after a period of service particularly with regard to the purity and moisture contents of the gas.

ii) The precise procedure to be adopted by maintenance personnel for handling equipment who are exposed to the products of arcing in SF6 gas, so as to ensure that they are not affected by possible irritants of the skin and respiratory system. Recommendations shall be furnished for suitable protective clothing, methods of disposal of circuit breaker cleaning utensils and other relevant matters.

h) A complete catalogue on operation analyzer satisfying all the requirements stipulated in this specification.

i) Curves supported by test data indicating the opening time under close open operation with combined variation of trip coil voltage and pneumatic/hydraulic pressure.

j) Detailed literature and schematic diagrams of switching mechanism for closing resistor showing the duration of insertion shall also be furnished along with calculations in respect of thermal rating of resistors for the duties.

k) All duty requirements shall be proved with the support of adequate test reports to be furnished along with the bid failing which the bid is likely to be rejected.

9.8 TEST REPORTS:

i) Two copies of acceptance test reports shall be furnished to the Purchaser. One copy will be returned, duly certified by the Purchaser and only there afterwards the material be dispatched.

ii) All records and routine test reports shall be maintained by the supplier at his works for periodic inspection by the Purchaser.

iii) All test reports of tests conducted during manufacture shall be maintained by the Supplier. These shall be produced for verification as and when requested for by the Purchaser.

10.0 PACKING AND FORWARDING & UNLOADING:

The equipment shall be packed in suitable crates so as to withstand handling during transit. The supplier shall be responsible for any damage to the equipment during transit, due to improper and inadequate packing and handling. The easily damageable materials shall be carefully packed and marked with the appropriate caution symbols. Wherever necessary, proper arrangement for lifting, such as lifting hooks etc., shall be provided. Any material found short inside the packing cases shall be supplied by the Supplier without any extra cost. Each consignment shall be accompanied by a detailed packing list containing the following information:

a) Name of the consignee.
b) Details of consignment.
c) Destination.
d) Total weight of consignment.
e) handling and unpacking instructions.
f) Bill of material indicating contents of each package and spare material.

The Supplier shall ensure that the packing list and bill of material are approved by the Purchaser before despatch.

The Supplier shall arrange for unloading the material/equipment at his cost.

11.0 QUANTITY AND DELIVERY REQUIREMENTS:

i) This is set out in the proforma given in Price Schedule and the prices shall be quoted accordingly.

ii) The scope of supply shall include supply of 25% extra-quantity of bolts, nuts, washers, split pins, cotter pins and such other small loose items free of cost.

12.0 SUPERVISION SERVICES:

The supplier is expected to provide services of qualified personnel for the supervision of erection, testing at site and commissioning the breakers supplied at free of cost.

13.0 MANDATORY SPARES & TOOLS:

Mandatory spares and tools as mentioned in Price Schedule shall also be included in the scope of supply and rates must be quoted accordingly.

14.0 GUARANTEED TECHNICAL PARTICULARS:

The Guaranteed Technical particulars for the equipment being supplied shall be provided with the Bid as specified in this technical specification at **Annexure-I.** The Bids received without guaranteed technical particulars shall be treated as Non-responsive.

The bidder must fill up all the points of GTP for offered item/s instead of indicating “refer drawing, or as per IS/IEC or as per test reports etc.”, the exact value/s must be filled in.

All the points other than GTP, which are asked to confirm in technical specifications must be submitted separately with the bid.

Annexure-I

GUARANTEED TECHNICAL PARTICULARS FOR 220 kV & 132 kV SF6 CIRCUIT BREAKERS & 33 kV VACUUM CIRCUIT BREAKERS

S. No	Description	Particulars		
		220 kV SF6 CBs	132 kV SF6 CBs	33 kV VCBs
1.	a) Maker's name country of manufacture. b) Manufacturer's type designation.			
2.	Applicable Technical Standards			
3.	a) Rated voltage (kV) b) Rated Frequency (Hz)			
4.	Number of Poles			
5.	Class (Outdoor/Indoor)			
6.	Rated normal current: a) Under site conditions (Amps) b) Rated (Amps)			
7.	Rated short circuit breaking current: a) R.M.S. value of A.C. component of rated short circuit current (kA) b) Percentage DC component c) Asymmetrical breaking current (including DC component) d) Certificate or report no. e) Oscillogram no.			
8.	Rated short circuit making current (kA)			
9.	First Pole to clear factor			
10.	Rated transient recovery voltage for terminal faults (kV peak)			
11.	Rated characteristics for short line faults.			
12.	Rated operating sequence .			
13.	Rated duration of short circuit			

	(sec.)			
14.	Rated out of phase breaking current (kA)			
15.	Opening time (ms)			
16.	Arcing time (ms) a) At 10% rated breaking current b) At 25% rated breaking current c) At 50% rated breaking current d) At 100% rated breaking current e) Maximum Arcing time at lowest fault currents			
17.	Break time (ms) a) At 10% rated breaking current b) At 25% rated breaking current c) At 50% rated breaking current d) At 100% rated breaking current e) Maximum break time at lowest fault current			
18.	Closing time (ms)			
19.	Maximum Pole discrepancy time: a) Opening (ms) b) Closing (ms)			
20.	Rated line charging breaking current (kA)			
21.	Maximum cable charging current a) On supply side b) On line side			
22.	Rated small inductive breaking current (kA)			
23.	Max. rise of temperature over ambient temperature for current rating under clause 6.			
24.	Interrupting capacity based on duty cycle as per clause 11. a) AC Component (kA) b) Percentage D.C. Component			
25.	Latching current (kA)			
26.	No of breaks in series per pole			
27.	Length of contact travel (mm)			
28.	Total length of break per pole			

	(mm)			
29.	Rate of contact travel : a) At tripping (meters/sec.) b) At closing (meters/sec.)			
30.	Type of devices, if any, usd to obtain uniform voltage distribution between breaks			
31.	Recovery voltage distribution between breaks in percent of rated voltage: a) Single line to ground fault b) Interruption of short lines c) Switching off an unloaded Transformers			
32.	Type of main contact			
33.	Type of arcing-contacts and/or arc control device			
34.	Material of contacts a) Main b) Arcing c) Auxiliary			
35.	Whether contacts are silver plated			
36.	Thickness of silver coating (mm)			
37.	Contact pressure (kg/sq. mm.)			
38.	Insulation level of the breaker : a) 1 minute power frequency withstand voltage (kV rms) b) Switching surge withstand test voltage (kV peak) c) Impulse withstand test voltage (kV peak) d) Max. dynamic p.f. over voltage withstand (kV peak)			
39.	Minimum clearance in Air (mm) a) Between Phases (live parts) b) Between live parts and earth c) Centre to centre distance between phases d) The safety boundaries during a breaking operation for circuit breakers with an external exhaust for ionized gases or flames			
40	Whether the circuit breaker is suitable for fixed trip operation or trip free operation and whether			

	it is provided with a lock-out device preventing closing of the breaker			
41.	Method of closing a) Normal b) Emergency			
42.	Type of closing mechanism			
43.	a) Normal voltage of closing b) Pick up range, (volts)			
44.	a) Normal voltage of closing b) Pick up range, (volts) a) Power at normal voltage of closing mechanism, (watts) b) Power at 85% of normal voltage, (watts)			
45.	Type of tripping mechanism			
46.	Normal voltage of tripping coils, (volts)			
47.	a) Power at normal voltage for tripping coils, (watts) b) Power at 70% normal voltage for tripping coils, (watts)			
48.	Arc duration at 100% (ms) Interruption capacity: a) Opening Arcing time No. of loops and time including resistor current duration (cycle) Resistor current duration, (cycle) Total length of the arc, (mm) Max. length of the arc, (mm) Total interrupting time measured from instant of trip coil energisation to arc extinction of resistor current (cycles). b) Closing time measured from instant of application of power to closing device upto arcing contacts touching, (cycles).			
49.	Critical current (current giving the longest arc when a break takes place) (kA)			
50.	a) Recovery voltage when circuit breakertested at 100% rated breaking capacity, (kV inst.) b) Rate of rise of re-striking voltage at breaking i) for 30% breaking capacity, (kV/Micro. sec.) ii) for 100% breaking capacity,			

	(kV/Micro. sec.) c) Maximum over voltage factor of the circuit breaker when switching off. i) Unloaded transformers. ii) Loaded transformer iii) Open circuited lines			
51.	When switching of synchronous systems: a) Max. current (kA) b) Max. contacts of 1 pole (kV)			
52.	No. of openings the circuit breaker is capable of performing without inspection, replacement of contacts or other main parts. a) at 50% rated current b) at 100% rated current c) at current corresponding to 50% rated breaking capacity d) at current corresponding to 100% rated breaking capacity			
53.	a) Weight of complete circuit breaker (kg.) b) Impact loading for foundation design, to include deed load plus impact value on opening at maximum interrupting ratings, in terms of equivalent static load, (kg.) c) Overall dimensions: Height (mm) Width (mm) Length (mm)			
54.	Porcelain: a) Make b) Type c) Descriptive pamphlet No. d) Weight (kg.) e) Transport dimensions, (mm) f) Height above floor, required to remove porcelain, (mm) g) Insulation class h) One minute dry power frequency withstand, kV (r.m.s.) i) 10 seconds wet power frequency withstand, kV (peak)			

	j) Flash over voltage, (kV) k) Full wave impulse withstand voltage kV (peak) l) Switching surge withstand voltage kV (peak) m) Corona discharge voltage, (kV r.m.s.) n) Nature of the dielectric o) Creepage distance, total protected (mm) p) Volume of insulating medium per porcelain, (liters) q) Permissible safe cantilever loading on installed porcelain (kg.m.)			
55.	Operating mechanism : a)Mechanically operated or pneumatically or hydraulically operated. b) For stored energy mechanism : 1. Spring charging motor i) Rating kW ii) Voltage V iii) Power frequency withstand voltage kV. iv) Time required for the motor to charge the spring fully (secs.) v) Power required at the normal control voltage to charge the spring - Watts. vi) Specification reference. 2. Spring closing/opening i) Number of springs ii) Type iii) Number of turns iv) Guage v) External diameter mm. vi) Stiffness vii) Material			

	viii) Force developed in full charged position. ix) Specification reference.			
56.	Pneumatic equipment for pneumatically operated breakers.(Not applicable) a) Type b) Manufacturers type designation c) Air Compressor i) Type ii) Make iii) Capacity in liters/min. d) **Compressor motor:** i) HP rating ii) Rated voltage and frequency iii) Limits of voltage and frequency variation for satisfactory operation of compressor motor iv) No. of phases v) Speed (r.p.m.) vi) Temp. of insulation. vii) Class of insulation. **e)** Safety, valve opens on receiver kg/sq.cm. f) Compressor start at kg/sq.cm. g) Compressor stops at kg/sq.cm. h) Alarm switch on receiver closes at kg/sq.cm. i) Lockout switch on receiver operates at i) kg/sq.cm. for closing ii) kg/sq.cm. for opening iii) kg/sq.cm. for auto re-closure duty j) Time of air compressor to charge the reservoirs: i) From atmospheric to pressure indicated in (g) above, (minutes)			

	ii) From pressure indicated in (g) above to that in (h) above (minutes)			
57.	Rated pressure of SF6 gas in the circuit breaker (kg/sq.cm)			
58.	Rated pressure of SF6 gas in the gas cylinders (kg./sq.cm.)			
59.	Quantity of SF6 gas required per single pole unit (kg.) (3 pole unit for 145 kV)			
60.	Quantity of SF6 gas per cylinder (kg.)			
61.	Weight of empty cylinder (kg.)			
62.	Quantity of absorbent required per pole (kg.)			
63.	Recommended interval for renewal of absorbent in case of outdoor circuit breakers operating in tropical conditions.			
64.	Chemical composition of absorbent			
65.	Quantity of absorbent covered in the scope of supply (			
66.	Limits of gas pressure for pressure operation of circuit breaker (kg./sq.cm.)			
67.	Pressure and temperature at which the temperature compensated gas pressure switch a) alarm (kg./sq.cm., deg.C) b) Cut off (kg/sq.cm. deg.C)			
68.	Name of SF6 supplier and country of origin			
69.	Quantity of SF6 gas supplied for a) Actual use in breakers (kg.) b) As spare (kg.)			
70.	Chemical composition of gas: a) Qty. of air by weight (ppm) b) Qty. of H20 by weight (ppm) c) Qty. of CF4 by weight (ppm) d) Qty of free acid by weight (ppm)			
72.	No. of auxiliary contacts provided a) Those close when breaker is closed. b) Those open when breaker is closed. c) Those adjustable with respect to the position of main			

	contacts. d) Continuous rating of contacts. e) Breaking capacity of contacts.			
73.	Whether the equipment covered by this Bid have been fully type tested and if so, whether the copies of the type test cert. enclosed to the bid offer.			

Name of the firm .....

Signature of Bidder .....

Designation & Seal .....

Date .....

The 132kv and 220kv circuit breakers shall be provided with pressure to current transducers of (4-20) mA to integrate the same to SAS for SF6 gas pressure in HMI

A C R O N Y M S

MVA	Mega Volt Ampere
kVA	Kilo Volt Ampere
kV	Kilo Volt
Cu. mt	Cubic meter
Deg. C	Degrees Centigrade
mm.	Milli meter
cm.	Centi meter
Dm.	Deci meter
km.	Kilo meter
g.	Grams
kg.	Kilo gram newton
V	Volts
Hz	Hertz
DC	Direct Current
AC	Alternate Current
ONAN	Oil Natural Air Natural
ONAF	Oil Natural Air Forced
HV	High Voltage
IV	Intermediary Voltage
Db	Decibel
KA	Kilo Ampers
sec.	Seconds
Amps.	Amperes
kVp.	Kilo volts peak
kV. Rms.	Kilo volts root mean square
KW	Kilo watt
Max.	Maximum
Min.	Minimum
UTS	Ultimate Tensile Strenth
QAP	Quality Assurance Plan
ACSR	Aluminium Conductor Steel Reinforced
AAAC	All Aluminium Alloy Conductor
HP	Horse Power
ppm.	Parts Per Million

OLTC	On Load Tap Change
WTI	Winding Temperature Indicator
OTI	Oil Temperature Indicator
Torr	Torr.
LOI	Letter of Intent
CIF	Cost including Insurance & Freight
CIP	Carriage Insurance Paid
SF6	Sulphur Hexa Flouride
TRV	Transient Recovery Voltage
CTD	Capacitor Tripping Device
ICU	Individual Compressor Unit
NO	Normally Open
NC	Normally Close

sq.mm Al. Cable	Square millimeter Aluminium cable
sq.mm Cu. Cable	Square millimeter Copper cable
MS Flat	Mild Steel Flat
C	Close
O	Open
HRC	High Rupturing Capacity
PVC	Poly Vinyle Chloride
p.f	.Power factor
CT	Current transformer
NCT	Natural Current Transformer
TEFC	Totally enclosed fan cooled
P.U.	Per Unit
ISO	Indian Standard Organisation
CB	Circuit Breaker
Auto	Automatic
kg./Sq.mm	Kilogram per square centimeter
T	Metric Tonn
Kn	Kilo Newtons